

Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline Clinical Insights

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Prostate cancer is a major cause of disease burden, with an estimated 313,780 new cases (the highest incidence in males aside from skin cancer, comprising 30% of all new cases) and approximately 33,363 deaths (the second highest cause of mortality in males and constituting 11% of all cancer-related deaths) in the United States in 2025.¹ As with many other cancers, somatic genomic testing has both prognostic and predictive capabilities and may affect treatment decision making for patients with metastatic prostate cancer.² Germline genetic testing may also provide valuable insights for treatment, with the added potential to offer additional hereditary cancer risk information requiring genetic counseling or pretest genetic education because of cancer risk relevance to patients and blood relatives including and beyond prostate cancer (eg, breast, pancreas, colon, endometrial, etc).^{2,3} To provide guidance on utility and implementation of germline and somatic genomic tests in patients with metastatic prostate cancer, ASCO published a guideline (see Fig 1 for recommendations and Fig 2 for the algorithm) to better inform clinicians, patients, and other stakeholders.⁴ That guideline addresses the following questions: (1) Who should receive germline testing? (2) Who should receive somatic testing? (3) Who should receive sequential somatic testing with next-generation sequencing technologies? (4) What are the strengths and weaknesses of primary tumor archival tissue versus fresh metastatic biopsy tissue versus circulating tumor DNA (ctDNA) testing for somatic testing? (5) What are the key therapeutic implications of germline or somatic testing for single-gene genetic variants (eg, *BRCA1*, *BRCA2*)? (6) What are the key prognostic implications of germline or somatic testing? This article provides additional considerations and nuances that may impact and inform the implementation of germline and somatic testing in metastatic prostate cancer.

WHAT ARE THE KEY INDICATIONS FOR GERMLINE VERSUS SOMATIC GENOMIC TESTING IN METASTATIC PROSTATE CANCER, AND HOW DO THESE TESTS INFORM TREATMENT SELECTION?

Current guidelines provide a comprehensive framework for germline and somatic genomic testing in metastatic prostate cancer, emphasizing its role in identifying patients eligible for targeted therapy and informing hereditary cancer risk assessment for patients and their families.^{2,3}

Indications for germline testing include metastatic disease diagnosis, high-risk or very high-risk disease features—such as advanced T-stage, elevated prostate-specific antigen, or higher Gleason score/Grade Group—and specific ancestry considerations, including Ashkenazi Jewish descent. Recent updates have expanded testing criteria to include early-onset disease (≤ 55 years) and biochemical recurrence following prior high-risk disease. Family history patterns indicative of hereditary cancer syndromes, including hereditary breast and ovarian cancer syndrome, Lynch syndrome, or classical hereditary prostate cancer, further inform testing decisions.³ Additionally, germline and somatic testing play a critical role in guiding targeted therapy selection, particularly for poly(ADP-ribose) polymerase (PARP) inhibitors and immunotherapies, based on the identification of pathogenic or likely pathogenic variants (P/LPVs) in DNA repair genes or DNA mismatch repair deficiency status.^{2,5-10}

WHAT STRATEGIES CAN CLINICIANS USE TO ADDRESS THE CHALLENGES OF SEQUENTIAL SOMATIC TESTING IN PATIENTS WITH METASTATIC PROSTATE CANCER?

Sequential somatic testing poses significant challenges in metastatic prostate cancer, particularly when initial results are inconclusive or negative. Retesting with metastatic biopsies or

Recommendation	Evidence Quality	Strength
1. All patients with metastatic prostate cancer should undergo germline genetic testing with next-generation sequencing technologies.	H	S
2. Those patients with metastatic prostate cancer (both CSPC and CRPC) who are being considered for biomarker-directed systemic treatment should undergo somatic testing with next-generation sequencing technologies.	H	S
<i>Practical information for Recommendation 2:</i> While there are no current FDA-approved biomarker directed treatments following somatic testing for mCSPC, somatic testing may be warranted in the presence of high-volume disease or where there is a high likelihood the patient's disease will progress to CRPC, where the patient is a candidate for future treatment with a biomarker directed therapy (PARP inhibitor or checkpoint inhibitor).		
3. The panel recommends that sequential somatic testing may be offered when there has been a meaningful change in the patient's status or treatment plan, especially in cases where prior tests were negative or uninformative (eg, insufficient or low tumor content)	M	W
4. Archival tissue samples are preferred in initial testing. ctDNA is preferred when there is no accessible metastatic site to biopsy or for sequential testing. In the setting of minimal disease burden associated with low ctDNA fraction, metastatic biopsy is preferred.	L	W
5. Patients with pathogenic germline variants or somatic alterations in <i>BRCA1</i> and <i>BRCA2</i> demonstrate poorer outcomes but are candidates for treatment with PARP inhibitor monotherapy, PARP inhibitor with ARPI combination therapy, and platinum-based agents	H	S
6. Treatment recommendations should not be made based on prognostic only biomarkers. However, they may be considered for directing patients to clinical trials. Germline information may still be important for patient counseling, informing hereditary risk for patients and families.	H	S

FIG 1. Summary of germline and somatic genomic testing for metastatic prostate cancer recommendations. ARPI, androgen receptor pathway inhibitor; CRPC, castration-resistant prostate cancer; CSPC, castration-sensitive prostate cancer; ctDNA, circulating tumor DNA; FDA, US Food and Drug Administration; H, high; L, low; M, moderate; mCSPC, metastatic castration-sensitive prostate cancer; NGS, next generation sequencing; PARP, poly(ADP-ribose) polymerase; PSA, prostate-specific antigen; S, strong; W, weak.

ctDNA provides critical opportunities to uncover actionable insights, particularly during disease progression.^{11–14} Metastatic biopsies offer a direct evaluation of tumor evolution, enabling the identification of new mutations with therapeutic relevance, such as homologous recombination deficiencies or microsatellite instability. However, the invasive

nature of these procedures and the limited accessibility of biopsy-amenable metastatic sites may restrict their feasibility in some patients. By contrast, ctDNA represents a minimally invasive alternative that captures tumor heterogeneity in real time, facilitating the detection of emerging genomic alterations and mechanisms of therapeutic

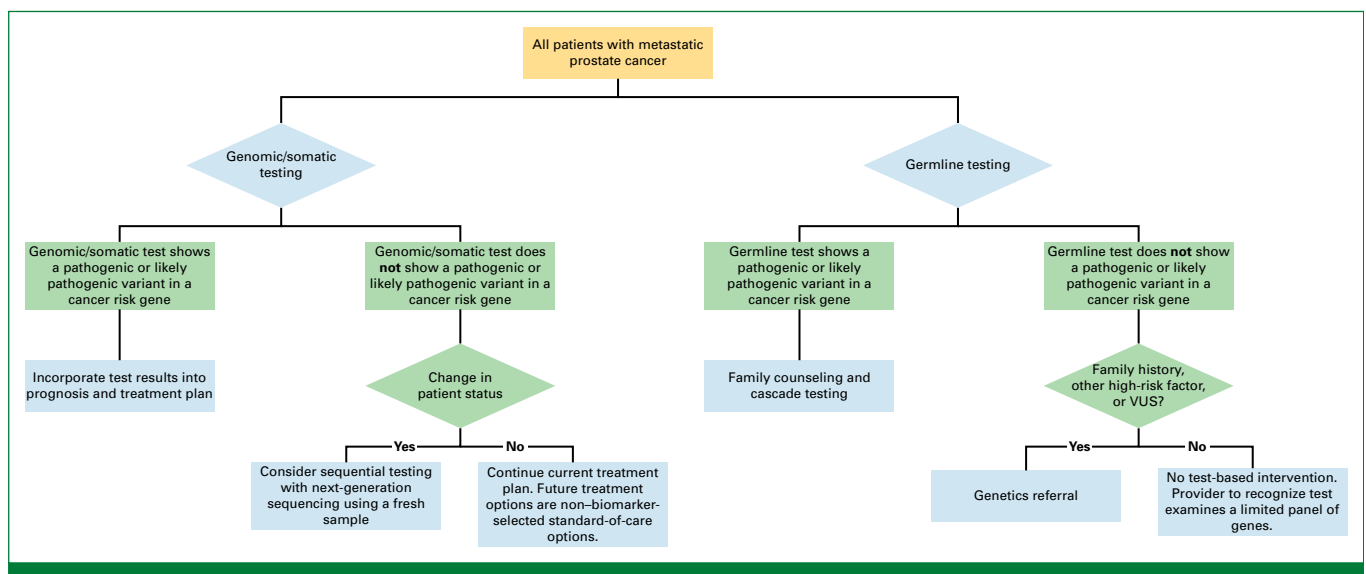


FIG 2. Germline and somatic genomic testing for metastatic prostate cancer. VUS, variants of uncertain significance.

resistance. Notably, retesting at critical times in the clinical course, such as disease progression to castration resistance, can identify new actionable findings in up to 11% of cases, highlighting its value in guiding treatment decisions.¹³ The successful integration of sequential testing into clinical practice necessitates close collaboration among oncologists, geneticists, and pathologists to optimize the selection of samples, timing of testing, and interpretation of results, thereby ensuring personalized and adaptive therapeutic strategies for patients.

HOW CAN CLINICIANS BALANCE THE BENEFITS OF GENOMIC TESTING WITH THE POTENTIAL PSYCHOLOGICAL IMPACT ON PATIENTS AND FAMILIES, ESPECIALLY REGARDING HEREDITARY CANCER RISK?

ASCO endorses the need for patients to make informed decisions prior to proceeding with germline testing.^{4,15} Furthermore, somatic testing may uncover potential germline findings, and therefore, informed consent about this potential result is also important to discuss with patients along with the need for possible confirmatory germline testing for genetic alterations suspected to be germline.

The gold standard for patients to make an informed decision for germline testing is through genetic counseling. Genetic counselors address the benefits, risks, and options for germline testing along with addressing the familial and psychological impact of hereditary cancer genetic testing.¹⁶ However, there is a relative scarcity of genetic counselors due to expanded indications for prostate cancer germline testing.^{2,3} Therefore, there has arisen a need for healthcare clinicians to offer germline testing with alternate delivery models such as point-of-care strategies to expand access to germline testing in a timely manner.^{17,18} These strategies necessitate tools to provide pretest genetic education that address the benefits, risks, and psychological impact of germline testing.¹⁹ Pretest educational materials also need to include information that cascade testing may be recommended if a P/LPV is identified in a patient. Videos, webtools, and print materials that are vetted by cancer genetics experts are key to ensure the accuracy of information given to patients about germline testing.²⁰⁻²³ An example of an online patient-driven webtool for germline pretest genetic education tested in a randomized trial is the TARGET tool.^{21,24}

WHAT ARE THE THERAPEUTIC IMPLICATIONS OF IDENTIFYING DNA REPAIR GENE MUTATIONS (eg, *BRCA1/2*) IN METASTATIC PROSTATE CANCER?

Multiple PARP inhibitors have received US Food and Drug Administration approval for treating patients with metastatic prostate cancer harboring DNA repair gene mutations, supported by clinical trials demonstrating improved outcomes.^{2,5-10} Consequently, identifying actionable DNA repair mutations through germline and somatic testing is essential for guiding targeted therapy. For patients with no prior docetaxel or no prior novel hormonal therapy,

combinations of niraparib + abiraterone and olaparib + abiraterone are given strong recommendations for patients with *BRCA* mutations, while talazoparib + enzalutamide combination is recommended for patients with mutations in a number of DNA repair genes including *BRCA1*, *BRCA2*, *ATM*, *ATR*, *CDK12*, *CHEK2*, *FANCA*, *MLH1*, *MRE11A*, *NBN*, *PALB2*, or *RAD51C*.² For patients treated with prior novel hormonal therapy and no prior docetaxel, recommendations are strong for olaparib or rucaparib for patients with *BRCA* mutations.² Additional options for PARP inhibitors include niraparib abiraterone for patients with *BRCA* mutations, olaparib abiraterone for patients with mutations in *BRCA1/2*, *ATM*, *BARD1*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *RAD51B*, *RAD51C*, *RAD51D*, or *RAD54L*, and talazoparib with enzalutamide for patients with mutations in *BRCA1*, *BRCA2*, *ATM*, *ATR*, *CDK12*, *CHEK2*, *FANCA*, *MLH1*, *MRE11A*, *NBN*, *PALB2*, or *RAD51C*.² For patients treated with prior docetaxel but no prior novel hormonal therapy, options for PARP inhibitors include niraparib + abiraterone or olaparib + abiraterone for patients with *BRCA* mutations or talazoparib + enzalutamide for patients with mutations in *BRCA1*, *BRCA2*, *ATM*, *ATR*, *CDK12*, *CHEK2*, *FANCA*, *MLH1*, *MRE11A*, *NBN*, *PALB2*, or *RAD51C*.² For patients whose disease has progressed on prior docetaxel and prior novel hormonal agents, olaparib is strongly recommended for patients with *BRCA* mutations and is an option for patients with mutations in *ATM*, *BARD1*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *RAD51B*, *RAD51C*, *RAD51D*, or *RAD54L*.² Rucaparib is also an option for patients with *BRCA* mutations.² As can be seen, there is variability in recommendations necessitating further study of clinical outcomes to PARP inhibitors based on genetic alterations in DNA repair genes.

WHAT ARE THE LIMITATIONS OF CURRENT GERMLINE AND SOMATIC TESTS IN METASTATIC PROSTATE CANCER, AND HOW CAN CLINICIANS NAVIGATE UNCERTAINTIES SUCH AS VARIANTS OF UNCERTAIN SIGNIFICANCE?

There are many considerations regarding selection, interpretation, and application of germline and somatic tests. Germline panels can have variability in the genes included on multigene panels and variant classification criteria across different clinical genetic laboratories. Furthermore, reporting of variants of uncertain significance (VUS) poses a clinical challenge to clinicians who order genetic tests. VUS do not impact clinical management at the time of reporting, but a fraction of VUS can be reclassified over time to P/LPVs which can be clinically actionable.²⁵ Since approximately one-third of males who undergo multigene panel testing for prostate cancer will have VUS identified, especially racial and/or ethnic minority populations,²⁵⁻²⁸ there is a need for ordering clinicians to ensure that patients understand the possibility of having a VUS reported and long-term plans if reclassification occurs. The ClinVar database from the National Institutes of Health²⁹ contains a repository of variant classifications from various laboratories; however, not all laboratories contribute to ClinVar. There can be discrepancies between labs regarding the classification of a unique

variant which can pose clinical challenges regarding actionability. Ordering clinicians are therefore urged to order germline testing from experienced genetic testing laboratories with robust variant classification programs.¹⁹

Current limitations of somatic testing include variability in findings based on sample type (formalin-fixed paraffin-embedded tissue or plasma cell-free DNA), variability in tumor fraction, potential of clonal hematopoiesis and false-positive results, and sample preparation leading to artifacts. Additional issues include lack of standardization of somatic test reporting which can impact identification of actionable variants. A significant amount of education is needed to inform which patients to test and when to test during the course of care, along with policies to ensure standardization of genomic test reports.³⁰

HOW DO GERMLINE TESTING RESULTS INFLUENCE FAMILY COUNSELING AND DECISIONS REGARDING CASCADE TESTING?

Individuals with P/LPVs on germline testing should have a discussion that these variants are heritable and, therefore, blood relatives are recommended to have germline

testing.^{31,32} Blood relatives should meet with a genetics professional to have this discussion for their own germline testing. P/LPVs on germline testing can provide key cancer risk information for patients about prostate cancer treatment, screening (for people without a diagnosis of prostate cancer) but also provide information about additional cancer risks relevant to people with or without prostate cancer. Cancer risks in families for people who inherit P/LPVs in cancer genes are also informative for blood relatives and help guide recommendations for cancer screening and risk reduction.³

The integration of germline and somatic genomic testing into metastatic prostate cancer management marks a pivotal advancement in precision oncology. These strategies refine treatment selection and provide crucial insights into hereditary cancer risk, facilitating proactive interventions. Effective implementation requires seamless collaboration among oncologists, genetic counselors, and multidisciplinary teams. Future research should enhance sequencing technologies, address disparities in genetic testing access, and expand biomarker-driven therapeutic strategies to further personalize care and improve outcomes in metastatic prostate cancer.

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DISCLAIMER

This clinical insights article is derived from recommendations in Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline. This document is based on an ASCO Guideline and is not intended to substitute for the independent professional judgment of the treating physician. Practice guidelines do not account for individual variation among patients. This article does not purport to suggest any particular course of medical treatment. Use of the guideline and this article are voluntary. Please refer to the complete guideline to understand the full recommendations.

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